

Natural-Tiling Candidate and Properness Certification Specification

Stage 10A: Exact Scientific Symmetry Action, Multidimensional Eligibility, and Ambiguity Preservation

mdstats

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1 Purpose and stage boundary

Stage 10A certifies whether a caller-proposed Stage-9 periodic cell complex is an eligible **natural-tiling candidate** under the currently available finite scientific evidence. It also proves or disproves properness relative to the full

exact automorphism group of one immutable `PeriodicNetView`.

Runtime/API target:

`mdstats 0.19.24a0`

The stage introduces four persistent result layers:

`PeriodicCellComplexSymmetryAction`
`NaturalTilingCertification`
`NaturalTilingCandidate`
`NaturalTilingCatalog`

Stage 10A does **not** discover natural face sets, propagate local face sectors, split provisional tiles along non-face strong rings, resolve published crossing alternatives, refine the primitive-ring bound, or run the LTA end-to-end natural-tiling search. Those responsibilities remain Stages 10B–10D.

The first backend is therefore a certifier, not a tiling generator:

caller-proposed Stage-9 complex
+ complete exact net symmetry
+ bounded strength evidence
+ face/witness compatibility
+ exact partition certificate
-> Stage-10A candidate certification

2 Scientific definitions

2.1 Natural-tiling candidate

A candidate is one scientific translation-labelled periodic cell complex whose selected face rings, tile attaching maps, and partition evidence satisfy the active Stage-10A gates. Candidate identity is the identity of the scientific complex, not the identity of any auxiliary triangulation or tetrahedral mesh.

2.2 Properness

Let G_{view} be the exact periodic net fixed by the selected `PeriodicNetView`, and let $\mathcal{T}$ be the proposed periodic cell complex. The target condition is

$$\text{Aut}(\mathcal{T}) = \text{Aut}(G_{\text{view}}).$$

The tiling uses the net as its fixed 1-skeleton. Every automorphism of the tiling therefore restricts to an automorphism of that net:

$$\text{Aut}(\mathcal{T}) \subseteq \text{Aut}(G_{\text{view}}).$$

Stage 10A proves equality by checking the reverse inclusion: every representative of the complete exact net automorphism group must preserve all scientific face orbits and tile attaching maps modulo lattice translation and orientation.

This properness condition follows the natural-tiling framework of Blatov, Delgado-Friedrichs, O’Keeffe, and Proserpio [1]. The exact translation-labelled cell action and its finite group-composition verification are `mdstats`-specific constructions built on the periodic vector representation [2] and the exact periodic-net symmetry layer [3].

3 Source contract

The high-level certifier consumes mutually consistent instances of:

```

certify_natural_tiling_candidate(
    view: PeriodicNetView,
    discovery: PeriodicNetSymmetryDiscovery,
    ring_index: PrimitiveRingIndex,
    strength_catalog: RingStrengthCatalog | None,
    face_certificates: Sequence[FacePlacementCertificate],
    selected_witnesses: Sequence[FaceEmbeddingWitness],
    compatibility: FaceCompatibilityConstraintSystem | None,
    complex_: PeriodicCellComplex,
    partition_certificate: PeriodicPartitionCertificate | None,
    *,
    symmetry_resources: NaturalTilingSymmetryResources | None = None,
) -> NaturalTilingCandidate

```

The first backend requires:

- a Stage-7R `PeriodicNetSymmetryDiscovery`, not an arbitrary user-supplied subgroup;
- a source-bound `PrimitiveRingSymmetryIndex` inside that discovery result;
- one scientific face-orbit representative per primitive-ring key;
- face certificates and selected witnesses aligned with every Stage-9 face;
- exact view, topology, embedding, ring-catalog, complex, partition, and compatibility source identity; and
- a Stage-9 complex that has already passed its chain, shell, and quotient invariants.

An absent strength catalog, compatibility system, or partition certificate does not cause a false negative. It produces an explicit unresolved certification dimension.

4 Exact scientific symmetry action

4.1 Oriented cell image

The image of an oriented quotient face or tile is represented by

```

OrientedCellImage(
    target_cell_index: int,
    image_shift: tuple[int, int, int],
    orientation: Literal[-1, 1],
)

```

For a source cell representative c , the record means

$$g(c, \mathbf{0}) = s(c', \delta), \quad s \in \{-1, +1\}.$$

A translated occurrence obeys

$$g(c, \mathbf{n}) = s(c', A_g \mathbf{n} + \delta),$$

where $A_g \in \text{GL}(3, \mathbb{Z})$ is the lattice matrix of the normalized periodic automorphism.

4.2 Face action

Each scientific face is an oriented `FacePlacement`. The existing exact ring action maps its primitive-ring placement and cyclic parameterization. The face orientation changes by the orientation of that parameterization.

For source face f_i and chosen target representative f_j ,

$$gf_i = s_{g,i} t_{\delta_{g,i}} f_j.$$

The first backend requires one selected face orbit per ring key. If a mapped ring key is absent from the selected face set, the operation is certified as not preserving the proposed tiling.

Auxiliary `FaceEmbeddingWitness` triangulations do not participate in this action. A symmetry-related face may be certified by a different admissible mesh witness without changing the scientific tiling.

4.3 Tile attaching-map action

A tile shell is an integer face chain

$$\partial_3 T_a = \sum_k c_k (f_{i_k}, \mathbf{n}_k), \quad c_k \in \{-1, +1\}.$$

Applying g gives

$$g(\partial_3 T_a) = \sum_k c_k s_{g, i_k} (f_{j_k}, A_g \mathbf{n}_k + \delta_{g, i_k}).$$

The resulting exact chain must equal a translated, optionally orientation-reversed target shell:

$$g(\partial_3 T_a) = s_{g,a} t_{\eta_{g,a}} \partial_3 T_b.$$

No geometric centroid, floating tolerance, auxiliary tetrahedron, or tile label is used. `PeriodicTileShell.label` remains non-authoritative metadata; scientific tile identity is determined by its translation-labelled attaching map.

4.4 Complete finite group-action check

Normalized periodic automorphism representatives compose up to one common lattice translation. Let

$$g_o g_i = t_{\mathbf{c}(o,i)} g_d,$$

where g_d is the stored normalized representative and $\mathbf{c}(o,i)$ is `composition_translation_table[o][i]`. For every face and tile image, Stage 10A verifies

$$\delta_d = A_o \delta_i + \delta_o - \mathbf{c}(o,i),$$

with target indices and orientation signs composed exactly. Failure is an internal invariant error, not an unresolved scientific result.

5 Public symmetry API

```
@dataclass(frozen=True)
class NaturalTilingSymmetryResources:
    max_operation_face_images: int = ...
    max_operation_tile_images: int = ...
    max_composition_checks: int = ...
```

All limits are preflighted transactionally.

```
build_periodic_cell_complex_symmetry_action(
    complex_: PeriodicCellComplex,
    symmetry: PeriodicNetSymmetry,
    ring_symmetry: PrimitiveRingSymmetryIndex,
    *,
    resources: NaturalTilingSymmetryResources | None = None,
) -> PeriodicCellComplexSymmetryAction
```

The result stores one dense operation record per exact group representative:

```
CellComplexSymmetryOperation(
    operation_index: int,
    status: PRESERVED | NOT_PRESERVED,
    face_images: tuple[OrientedCellImage, ...],
    tile_images: tuple[OrientedCellImage, ...],
    reason: str | None,
)
```

A missing exact face or tile image is a proof of non-invariance. It is not a resource-truncation state. Resource exhaustion raises `NaturalTilingResourceError` before publishing a partial action.

6 Multidimensional certification

6.1 Independent state enum

`CertificationState` = CERTIFIED | REJECTED | UNRESOLVED

The persistent certification record is

```
NaturalTilingCertification(
    primitive_ring_bound: int,
    primitive_complete: CertificationState,
    symmetry_complete: CertificationState,
    strength_complete: CertificationState,
    embedding_complete: CertificationState,
    compatibility_complete: CertificationState,
    cell_complex_valid: CertificationState,
    partition_certified: CertificationState,
    properness: PropernessStatus,
    resource_truncations: tuple[str, ...],
    unresolved_assumptions: tuple[str, ...],
    rejection_reasons: tuple[str, ...],
)
```

The dimensions have the following meaning.

Dimension	CERTIFIED condition	REJECTED condition	UNRESOLVED examples
primitive	lower-closed primitive catalog, no truncation, complete through every selected face size	reserved	incomplete bound or source truncation
symmetry	complete Stage-7R discovery and source-bound ring action	reserved	unsupported future discovery domain
strength	every selected face is <code>STRONG_IN_DOMAIN</code> in its declared finite domain	at least one selected face is <code>WEAK_CERTIFIED</code>	missing result, truncated domain, incomplete source
embedding	every selected face has the chosen certified admissible disk witness	reserved	missing or non-admissible witness
compatibility	selected witness tuple violates no finite constraint and retains no unresolved constraint	selected tuple activates a hard forbidden constraint	absent system or active unresolved constraint
cell complex	Stage-9 scientific object exists and validates	reserved	future conditional complex backend

Dimension	CERTIFIED condition	REJECTED condition	UNRESOLVED examples
partition	exact Stage-9 certificate belongs to this complex	reserved	certificate absent

STRONG_IN_DOMAIN remains explicitly bounded. Stage 10A does not promote it to an unbounded global strong-ring theorem.

6.2 Properness state

```
PropernessStatus =
    CERTIFIED_PROPER
    | CERTIFIED_IMPROPER
    | UNRESOLVED
```

For the current first backend:

- a complete discovery plus a preserved scientific action gives `CERTIFIED_PROPER`;
- a complete discovery plus any non-preserving operation gives `CERTIFIED_IMPROPER`; and
- future unsupported or incomplete symmetry backends must use `UNRESOLVED`.

6.3 Aggregate eligibility

```
CandidateEligibility = ELIGIBLE | INELIGIBLE | UNRESOLVED
```

The aggregation rule is deterministic:

1. Any rejected dimension, certified improperness, or rejection reason gives `INELIGIBLE`.
2. All dimensions certified, properness certified, and no unresolved/truncation record gives `ELIGIBLE`.
3. Every other combination gives `UNRESOLVED`.

The aggregate never replaces the independent dimensions.

7 Candidate identity and evidence identity

```
NaturalTilingCandidate(
    periodic_net_view_digest: str,
    topology_graph_digest: str,
    primitive_ring_catalog_digest: str,
    periodic_cell_complex_digest: str,
    selected_ring_keys: tuple[PrimitiveRingKey, ...],
    certification: NaturalTilingCertification,
    symmetry_action_digest: str | None,
    partition_certificate_digest: str | None,
    strength_catalog_digest: str | None,
    compatibility_system_digest: str | None,
    digest: str,
    evidence_digest: str,
)
```

Two digests are intentional.

- `digest` is the scientific candidate identity. It depends on the immutable source identities, the Stage-9 complex, and the selected ring keys.
- `evidence_digest` includes certification states and evidence digests.

Thus two different exact tetrahedralizations or admissible face meshes can certify the same scientific tiling without creating duplicate candidates.

8 Catalog and ambiguity

NaturalTilingOutcomeKind = NONE | UNIQUE | MULTIPLE

```
NaturalTilingCatalog(
    periodic_net_view_digest: str,
    primitive_ring_catalog_digest: str,
    candidates: tuple[NaturalTilingCandidate, ...],
    outcome: NaturalTilingOutcome,
    essential_ring_keys: tuple[PrimitiveRingKey, ...],
)
```

Only ELIGIBLE candidates contribute to the outcome:

$$N_{\text{eligible}} = 0 \Rightarrow \text{NONE},$$

$$N_{\text{eligible}} = 1 \Rightarrow \text{UNIQUE},$$

$$N_{\text{eligible}} > 1 \Rightarrow \text{MULTIPLE}.$$

Unresolved candidates remain stored even when the eligible outcome is NONE. Enumeration order is never a scientific tie-breaker. Duplicate scientific digests are collapsed deterministically; conflicting eligible and rejected evidence for one scientific identity is rejected as inconsistent input.

An empty catalog is permitted when the caller supplies the view and ring-catalog digests explicitly.

9 Essential rings

A primitive-ring orbit is called essential only when it is used as a face by an eligible candidate. The catalog stores the union

$$\mathcal{R}_{\text{essential}} = \bigcup_{\mathcal{T} \in \mathcal{C}_{\text{eligible}}} \mathcal{R}_{\text{face}}(\mathcal{T}).$$

Unresolved or rejected candidates do not create essential rings.

10 Persistence and replay

Canonical schemas:

```
mdstats.cell-complex-symmetry-action.v1
mdstats.natural-tiling-candidate.v1
mdstats.natural-tiling-catalog.v1
```

All records use canonical JSON and SHA-256 digests.

- `PeriodicCellComplexSymmetryAction.from_dict(...)` rebuilds the exact action from the supplied complex, symmetry, and ring-symmetry sources and rejects any altered image table.
- Candidate and catalog deserialization revalidates canonical identity, evidence, multiplicity, and essential-ring invariants.
- Dense local indices remain bound to source digests.

11 Resource and failure policy

The scientific result is never partially published after a resource failure.

Condition	Result
face/tile image preflight exceeds declared limit	raise <code>NaturalTilingResourceError</code>
group-composition preflight exceeds declared limit	raise <code>NaturalTilingResourceError</code>
exact mapped face absent	operation <code>NOT_PRESERVED</code>
exact mapped tile shell absent	operation <code>NOT_PRESERVED</code>
ambiguous duplicate target tile orbit	raise <code>NaturalTilingInputError</code>
altered serialized action	raise <code>NaturalTilingSerializationError</code>
missing optional strength/compatibility/partition evidence	candidate <code>UNRESOLVED</code>
certified weak face or certified improper symmetry	candidate <code>INELIGIBLE</code>

12 Validation fixtures

The focused first-backend fixture is the simple-cubic net with one vertex orbit, three translation-labelled edge orbits, three square primitive-ring face orbits, and one cubic tile orbit. Exact symmetry discovery produces 48 normalized operations.

The Stage-10A gate verifies:

- 48 preserved scientific operation records;
- three face images and one tile image per operation;
- orientation-reversing tile images where required;
- exactly $48^2(3 + 1) = 9216$ face/tile group-composition checks;
- independence from auxiliary tetrahedral mesh identity;
- explicit non-preservation when a scientific image is absent;
- transactional resource failure;
- eligible/proper/unique certification;
- unresolved missing compatibility or partition evidence;
- rejection of a certified weak selected face;
- scientific deduplication across auxiliary evidence;
- explicit multiple eligible identities;
- empty `NONE` catalogs;
- serialization replay; and
- digest/tamper rejection.

13 Explicit non-responsibilities

Stage 10A does not:

- enumerate face selections;
- infer tile shells from local face sectors;
- split cells along admissible non-face strong rings;
- implement crossing-ring alternatives;
- prove global unbounded ring strength;
- perform primitive-bound refinement and downstream rebuild;
- certify the LTA $[4^6]$, $[4^6.6^8]$, and $[4^{12}.6^8.8^6]$ tile set;
- compute tile geometry, cages, or portals; or
- require auxiliary meshes to be symmetric as combinatorial triangulations.

14 References

1. V. A. Blatov, O. Delgado-Friedrichs, M. O’Keeffe, and D. M. Proserpio, *Three-Periodic Nets and Tilings: Natural Tilings for Nets*, Acta Crystallographica Section A **63**, 418–425 (2007), DOI: 10.1107/S0108767307038287.
2. S. J. Chung, Th. Hahn, and W. E. Klee, *Nomenclature and Generation of Three-Periodic Nets: The Vector Method*, Acta Crystallographica Section A **40**, 42–50 (1984), DOI: 10.1107/S0108767384000088.
3. O. Delgado-Friedrichs and M. O’Keeffe, *Identification of and Symmetry Computation for Crystal Nets*, Acta Crystallographica Section A **59**, 351–360 (2003), DOI: 10.1107/S0108767303012017.